POINTS OF A GROUPOID OVER A FIELD CAN BE TRANSPORTED BY AN ISOMORPHISM OF SOURCE FIBRES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-groupoids.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- Two source fibres of a groupoid over a field become isomorphic over a common field extension, transporting a prescribed pair of points (more-groupoids chapter, Lemma lemma-two-fibres, no canonical tag in the snapshot; no-tag pointer to the source label line, parts (1) and (2)). cited at proof lines 1213-1215. Its parts (1) and (2) give verbatim the field extension, the isomorphism phi with phi(r_1) = r_2, and the projections pr_0(r_1) = r, pr_0(r_2) = r'.

Source reference. Original lemma lemma-two-fibres: sources/stacks-more-groupoids/pinned-original-source.tex, source lines 761–761. Cited at proof lines 1213-1215: parts (1) and (2) give the field extension, the isomorphism phi carrying r_1 to r_2, and the projections to r and r'.

Lemma 0.2. *Let S be a scheme. Let (U, R, s, t, c) be a groupoid scheme over S. Assume $U = \mathrm{Spec}(k)$ with k a field. For any points $r, r' \in R$ there exists a field extension k'/k and points $r_1, r_2 \in R \times_{s,\mathrm{Spec}(k)} \mathrm{Spec}(k')$ and a diagram*

$$R \xleftarrow{pr_0} R \times_{s,\mathrm{Spec}(k)} \mathrm{Spec}(k') \xrightarrow{\varphi} R \times_{s,\mathrm{Spec}(k)} \mathrm{Spec}(k') \xrightarrow{pr_0} R$$

such that φ is an isomorphism of schemes over $\mathrm{Spec}(k')$, we have $\varphi(r_1) = r_2$, $pr_0(r_1) = r$, and $pr_0(r_2) = r'$.

Proof. This is a special case of Lemma lemma-two-fibres parts (1) and (2). □

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